

Damien Robert

Deep Learning for Remote Sensing & Environment



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I am a postdoctoral researcher in the [EcoVision](#) lab at the [University of Zurich](#), where I collaborate with [Jan Dirk Wegner](#) to design deep learning methods for remote sensing and environmental applications. Before joining UZH, I completed my PhD on “*Efficient Learning on Large-Scale 3D Point Clouds*” at [IGN](#) and [ENGIE](#), under the supervision of [Loic Landrieu](#) and [Bruno Vallet](#).

3D point clouds vegetation mapping species distribution remote sensing efficient ML

■ POSITIONS

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|-----------|--|
| 2024–now | Postdoctoral Researcher
EcoVision lab, University of Zurich · Zurich, Switzerland
Deep learning for remote sensing and environment
PI: Jan Dirk Wegner |
| 2020–2024 | PhD Student
LASTIG, IGN / ENGIE Lab CRIGEN · Paris, France
Efficient learning on large-scale 3D point clouds
Advisors: Loic Landrieu, Bruno Vallet |
| 2017–2020 | R&D Engineer
SIRADEL, ENGIE / ENGIE · Rennes, France
Deep learning on large-scale terrestrial/aerial, indoor/outdoor 3D and 2D data |
| 2017 | Co-Founder
Inspirama · Rennes, France
Website gathering book recommendations from inspiring people |
| 2015 | R&D Intern
Dassault Systèmes · Providence, RI, USA
Dimensionality reduction and dynamic system modeling |
| 2014 | R&D Intern
Dassault Systèmes · Providence, RI, USA
UX design |
| 2014 | R&D Intern
Dassault Systèmes · Providence, RI, USA
UX design |

■ EDUCATION

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|-----------|---|
| 2020–2024 | PhD, Computer Science
Université Gustave Eiffel · Paris, France
Efficient Learning on Large-Scale 3D Point Clouds |
| 2022 | Summer School
ICVSS · Sicily, Italy
Computer vision courses by world-renowned experts in academia and industry |
| 2017 | AI Fall School
CNRS · Lyon, France
Multi-disciplinary course for AI students and researchers |
| 2011–2015 | MSc, Engineering
École Centrale de Lyon · Lyon, France
Mathematics, Computer Science, Mechanics, Signal Processing, Automation |

■ AWARDS

- 2026 **Outstanding Reviewer** · [CVPR](#)
- 2025 **PhD Award accessit** · [AFRIF](#)
French Association for Pattern Recognition and Interpretation
- 2024 **Outstanding Reviewer** · [ECCV](#)
- 2022 **Best Paper Finalist** · [CVPR](#)
Top 0.4% of submissions

■ PUBLICATIONS

🗣️ 487 citations · h-index 7 · Google Scholar, Sep 2026  2022–2026

- 2026 **Toward a Foundation Model for Forest Point Clouds**
Y. Yue, S. Puliti, **D. Robert**, A. Topaloğlu, B. Xiang, M. Wielgosz, J. D. Wegner, R. Astrup, C. Rupprecht, K. Schindler
Under review
- From Machine Learning to Large-Scale EO Products: Best Practices for Making Maps**
G. Sialelli, R. Young, Y. Jiang, ..., **D. Robert**, et al. (13 authors)
Under review
- Bias Aware Benchmark for Species Distribution Modeling**
E. Arens, N. van Tiel, R. Zbinden, C. Vanalli, **D. Robert**, L. Drees, B. Kellenberger, L. Pellissier, J. D. Wegner, D. Tuia
[World Biodiversity Forum](#)
- LitePT: Lighter Yet Stronger Point Transformer**
Y. Yue, **D. Robert**, J. Wang, S. Hong, J. D. Wegner, C. Rupprecht, K. Schindler
[CVPR](#) [Highlight](#)
- EZ-SP: Fast and Lightweight Superpoint-Based 3D Segmentation**
L. Geist, L. Landrieu, **D. Robert**
[ICRA](#)
- FORMSPoT: Revealing fine-scale forest disturbances from nation-wide 1.5 m forest canopy height time series**
M. Schwartz, F. Fogel, N. Besic, **D. Robert**, L. Geist, J. Renaud, J. Monnet, C. Mosig, C. Vega, A. d'Aspremont, L. Landrieu, P. Ciaï
[Remote Sensing of Environment](#)
- 2025 **MT-GSR4B: Multi-Temporally Guided Super-Resolution for Above-Ground Biomass Estimation**
K. Karaman, V. S. F. Garnot, **D. Robert**, M. J. Santos, J. D. Wegner
[PFG – Journal of Photogrammetry, Remote Sensing and Geoinformation Science](#)
- SSL4Eco: A Global Seasonal Dataset for Geospatial Foundation Models in Ecology**
E. Plekhanova, **D. Robert**, J. Dollinger, E. Arens, P. Brun, J. D. Wegner, N. Zimmermann
[CVPR workshop EarthVision](#)
- GSR4B: Biomass Map Super-Resolution with Sentinel-1/2 Guidance**
K. Karaman, Y. Jiang, **D. Robert**, V. S. F. Garnot, M. J. Santos, J. D. Wegner
[ISPRS Geospatial Week](#) [Oral](#)
- Climplicit: Climatic Implicit Embeddings for Global Ecological Tasks**
J. Dollinger, **D. Robert**, E. Plekhanova, L. Drees, J. D. Wegner
[ICLR workshop Tackling Climate Change with Machine Learning](#)
- Lossy Neural Compression for Geospatial Analytics: A Review**
C. Gomes, I. Wittmann, **D. Robert**, et al. (27 authors)
[IEEE Geoscience and Remote Sensing Magazine](#)
- 2024 **Efficient Learning on Large-Scale 3D Point Clouds**
D. Robert
[PhD Thesis, Université Gustave Eiffel](#) [AFRIF PhD award accessit](#)
Jury: Sébastien Lefèvre, Cédric Demonceaux, Patrick Pérez, Siyu Tang, Duygu Ceylan, Loïc Landrieu, Bruno Vallet
- Scalable 3D Panoptic Segmentation as Superpoint Graph Clustering**
D. Robert, H. Raguet, L. Landrieu
[3DV](#) [Oral](#) · top 5.3% of submissions
- 2023 **Efficient 3D Semantic Segmentation with Superpoint Transformer**
D. Robert, H. Raguet, L. Landrieu

ICCV · 26.8% acceptance rate

2022

Learning Multi-View Aggregation In the Wild for Large-Scale 3D Semantic Segmentation

D. Robert, B. Vallet, L. Landrieu

CVPR **Best paper finalist** **Oral** · top 0.4% of submissions

■ COLLABORATION & SUPERVISION

Collaborators	Konrad Schindler ETH · Christian Rupprecht Oxford · Jan Dirk Wegner UZH · Loic Landrieu ENPC · Vivien Sainte Fare Garnot UZH · Lukas Drees UZH · Martin Schwartz LSCE · Elena Plekhanova WSL
PhD Students	Emilia Arens UZH with Jan Dirk Wegner · Johannes Dollinger UZH with Jan Dirk Wegner · Kaan Karaman UZH with Jan Dirk Wegner · Louis Geist ENPC with Loic Landrieu · Yuanwen Yue ETH with Konrad Schindler and Christian Rupprecht
Master Students	Valerio Schelbert WSL · Jackson Sunny École Polytechnique with SAMP
Advising	UpCircle AI Switzerland · École des Sciences Criminelles de Lausanne Switzerland · SAMP AI France · GEODIGITAL US

■ REVIEWING, CHAIRING & ORGANISING

 Reviewer  Area chair  Organising committee

2027	 CVPR workshop EarthVision
2026	 ECCV ·  CVPR
2025	 CVPR ·  CVPR workshop EarthVision
2024	 ECCV ·  ISPRS Journal of Photogrammetry and Remote Sensing ·  ICLR workshop ML4RS ·  CVPR workshop EarthVision
2023	 CVPR ·  CVPR workshop EarthVision
2022	 ISPRS Journal of Photogrammetry and Remote Sensing ·  CVPR workshop EarthVision

■ TEACHING

2026	Transformer · UZH · Lecture, Master 2
2025	AI4Good group projects · UZH · Labs, Master 2
2025	Transformer · UZH · Lecture, Master 2
2024	Transformer, NeRFs, and Diffusion · UZH · Lecture and labs, Master 2
2023	Deep Learning for Remote Sensing · ENSG-IGN · Lecture, Master 2
2022	3D Deep Learning for Remote Sensing · ISPRS'22 · Tutorial, Researchers
2022	3D Deep Learning, Torch-Points3D & DeepViewAgg · ENGIE CRIGEN lab · Tutorial, Researchers
2022	Deep Learning for Remote Sensing · ENSG-IGN · Lecture, Master 2
2020	Deep Learning for Computer Vision · École Polytechnique · Lecture, Master 1

■ OPEN-SOURCE SOFTWARE

★ 1055	 136	Superpoint Transformer ·  drprojects/superpoint_transformer
★ 406	 45	LitePT ·  prs-eth/LitePT
★ 237	 24	DeepViewAgg ·  drprojects/DeepViewAgg
★ 74	 8	Point Geometric Features ·  drprojects/point_geometric_features
★ 53	 3	NoRA ·  drprojects/nora
★ 15	 0	torch-graph-components ·  drprojects/torch-graph-components

■ SELECTED TALKS

 Invited  Conference  Poster  Tutorial  Interview

2026	 ESA-NASA workshop on AI Foundation Model for EO · Building Earth Observation Embedding Workflows with TerraTorch · Virtual
	 ESA Φ-lab · Geospatial foundation models for forest disturbance monitoring · Frascati, Italy

2025	🗨️ MPI of Animal Behavior · Research at EcoVision lab · Zurich, Switzerland 🗨️ Global Forest Watch · Geospatial foundation models for forest disturbance monitoring · Virtual 🗨️ NatureFinance · Research at EcoVision lab · Virtual 🗨️ NVIDIA Strategic Research Engagement · Superpoint Transformer and EcoVision research · Zurich, Switzerland 🗨️ CNES COMET TSI · Remote Sensing for Species Distribution Modeling · Virtual
2024	🗣️ 3D Data Academy · Interview by Florent Poux and tutorial for Superpoint Transformer · YouTube 🗨️ Google DeepMind · Research at EcoVision lab · Zurich, Switzerland 🎤 3DV'24 · Scalable 3D Panoptic Segmentation as Superpoint Graph Clustering · Davos, Switzerland
2023	🗨️ École des Ponts, IMAGINE lab · Efficient Learning on Large-Scale 3D Point Clouds · Paris, France 🗨️ National Land Survey of Finland (NLS) · IGN's research on Large-Scale 2D and 3D Learning · Paris, France 🗨️ ETH Zürich, Computer Vision and Geometry lab · Efficient Learning on Large-Scale 3D Point Clouds · Zurich, Switzerland 🗨️ ETH Zürich, Photogrammetry and Remote Sensing lab · Efficient Learning on Large-Scale 3D Point Clouds · Zurich, Switzerland 🗨️ Valeo.ai · Efficient 3D Semantic Segmentation with Superpoint Transformer · Paris, France
2022	🗣️ CV News · Interviewed by the CV News journal for its Best of CVPR'22 issue · New Orleans, US 🎤 CVPR'22 · Learning Multi-View Aggregation In the Wild for Large-Scale 3D Semantic Segmentation · New Orleans, US
45 talks given since 2021 — full list at drprojects.github.io/talks .	

■ SKILLS & LANGUAGES

Research	3D & geometry Large-scale 3D point clouds · LiDAR data · Superpoint-based learning · Graph-based learning and graph partitioning Efficient & scalable learning Efficient deep learning · Scalable model design · EO data representation and compression Learning paradigms Multimodal learning · Self-supervised learning · Representation learning Environment & Earth observation Remote sensing · Forest structure analysis · Species distribution modeling · Representation learning for macroecology			
Tools	 Python	 PyTorch	 PyTorch Lightning	 PyTorch Geometric
	 C++	 Hydra	 Weights & Biases	 Git
	 Linux	SLURM	 scikit-learn	 Plotly
	 Blender	LaTeX		
Languages	French Native	English Fluent	Spanish Intermediate	German Beginner
Abroad	Zurich, Switzerland 2024–now · Backpacking, Worldwide 2015–2016 · Providence, RI, United States 2014–2015			
Interests	Nature & forests · Boxing · Mountaineering · Skateboarding · Guitar · Painting · Reading · Board games			